

JEE MAIN REVISION MASTER: REDOX REACTIONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 71 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Redox Reactions - Core Application	71 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Redox Reactions - Core Application (Very High Yield — 71 PYQs)

Core Principle: Key problem pattern in Redox Reactions.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "In order to oxidise a mixture of 1 mole each of $\mathrm{FeC}_2\mathrm{O}_4$, $\mathrm{Fe}_2\mathrm{C}_2\mathrm{O}_4$ and $\mathrm{C}_2\mathrm{O}_4^{2-}$, KMnO_4 is required. The volume of KMnO_4 solution (0.1 M) required is x ml. The value of x is..." [Ans: B]
- [2026 Evening Shift] Q#1 (Numerical): "200 cc of $x \times 10^{-3} \text{ M}$ potassium dichromate is required to oxidise 750 cc of 0.6 M Mohr's salt solution in acidic medium. Here $x = \dots$ " [Ans: 375]
- [2026 Morning Shift] Q#2 (MCQ): "Given below are two statements: Statement (I): Oxidising power of halogens decreases in the order $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$, which is the base..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

In order to oxidise a mixture of 1 mole each of FeC_2O_4 , $\text{Fe}_2(\text{C}_2\text{O}_4)_3$, FeSO_4 and $\text{Fe}_2(\text{SO}_4)_3$ in acidic medium, the number of moles of KMnO_4 required is

- (A) 3
- (B) 2 **[Correct]**
- (C) 5
- (D) 7

Correct Answer: B

Solution Analysis: In acidic medium, KMnO_4 acts as a strong oxidising agent. So, we need to find which species in the given mixture can be oxidised further. The mixture contains 1 mole each of: FeC_2O_4 , $\text{Fe}_2(\text{C}_2\text{O}_4)_3$, FeSO_4 , $\text{Fe}_2(\text{SO}_4)_3$. Now let us examine each compound. 1. FeC_2O_4 This is ferrous oxalate, so it contains: Fe^{2+} : can be oxidised to Fe^{3+} ; $\text{C}_2\text{O}_4^{2-}$: can be oxidised to CO_2 . Oxidation of Fe^{2+} : $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + e^-$. So, 1 mole of Fe^{2+} requires 1 mole of KMnO_4 . Oxidation of $\text{C}_2\text{O}_4^{2-}$: $\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2 + 2e^-$. So, 1 mole of $\text{C}_2\text{O}_4^{2-}$ requires 1 mole of KMnO_4 . In FeC_2O_4 , there is 1 mole of Fe^{2+} and 1 mole of $\text{C}_2\text{O}_4^{2-}$. So, 1 mole of FeC_2O_4 requires 2 moles of KMnO_4 . 2. $\text{Fe}_2(\text{C}_2\text{O}_4)_3$ This contains 2 moles of Fe^{2+} and 3 moles of $\text{C}_2\text{O}_4^{2-}$. So, 2 moles of Fe^{2+} require 2 moles of KMnO_4 and 3 moles of $\text{C}_2\text{O}_4^{2-}$ require 3 moles of KMnO_4 . So, 1 mole of $\text{Fe}_2(\text{C}_2\text{O}_4)_3$ requires 5 moles of KMnO_4 . 3. FeSO_4 This contains 1 mole of Fe^{2+} . So, 1 mole of Fe^{2+} requires 1 mole of KMnO_4 . 4. $\text{Fe}_2(\text{SO}_4)_3$ This contains 2 moles of Fe^{3+} . It is already in its highest oxidation state and cannot be oxidised further. So, it requires 0 moles of KMnO_4 . Total moles of KMnO_4 required = 2 (from FeC_2O_4) + 5 (from $\text{Fe}_2(\text{C}_2\text{O}_4)_3$) + 1 (from FeSO_4) + 0 (from $\text{Fe}_2(\text{SO}_4)_3$) = 8 moles.